



ESTD : 1946

THE NATIONAL INSTITUTE OF ENGINEERING MYSORE – 8

(Autonomous Institution under VTU)

B.E - CSE

VII and VIII Semester

Department of Computer Science and Engineering

The National Institute of Engineering														
Scheme of Teaching & Examination														
Department: Computer Science and Engineering (BE in CS&E)														
B.E. 2023 Admitted batch														
VII SEMESTER														
Sl.No	Type of Course	Course Code	Course Title	Teach ing Depart ment (TD)	Question Paper setting Board (PSB)	Teaching Hrs/Week				Examination				Credits
						L	T	P	S	Duration in Hours	CIE Marks	SEE Marks	Total Marks	
1	PCC	BCS701	Cryptography & Network Security	CS	CS	4	0	0		03	50	50	100	4
2	IPCC	BCS702	Parallel Computing	CS	CS	3	0	2		03	50	50	100	4
3	PEC	BCS713x	Professional Elective Course (Group III)	CS	CS	3	0	0		03	50	50	100	3
4	OEC	BCS754x	Open Elective Course (Group II)	CS	CS	3	0	0		01	50	50	100	3
5	PROJ	BCS785	Major Project	CS	CS	0	0	12		03	100	100	200	6
6	NCMC	1BUHK704	Universal Human Values	MOOC		-	2	-		-	50	-	-	0
											350	300	600	20
Professional Elective Course														
BCS713A		Deep Learning							BCS713D		Big Data Analytics			
BCS713B		Natural Language Processing							BCS713E		Quantum Computing			
BCS713C		No SQL							BCS713F		Digital Forensics			

Code: BCS702**Course: Parallel Computing****Credits: 4****L:T:P - 3:0:2****SEE: 50% Marks****CIE: 50% Marks****SEE Hours: 3 hours****Max. Marks:100**

Prerequisites if any	Computer organization, C programming
Learning objectives	1. To explore the need for parallel programming 2. To familiarize MIMD systems, parallelism in MPI library, OpenMP pragma and directives

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Discuss basic concepts of Parallel Computing	Understanding
CO2	Apply OpenMP and MPI concepts to write parallel programs	Apply
CO3	Analyze different parallel computing architectures and parallel programming models to identify suitability for specific computational problems.	Analyze
CO4	Evaluate speedup, efficiency, and scalability trends of parallel programs by interpreting performance metrics.	Evaluate

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	3	1	-	-		-	-	2	-	-	-	3		-	-
CO2	3	3	3	2	2	-	2	-	2	2	2	3		3	2
CO3	3	3	2	3	2	-	1	-	2	2	2	3		2	2
CO4	3	3	2	3	2	-	1	2	2	2	2	3		2	2

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

Sl. No.	Modules	No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
Module – 1: Introduction to parallel programming, Parallel hardware and parallel software				
1.1	Classifications of parallel computers, SIMD systems, MIMD systems	2	-	-
1.2	Interconnection networks	2	-	-
1.3	Cache coherence	2	-	-
1.4	Coordinating the processes/threads,	1	-	-
1.5	Shared-memory, Distributed-memory	1	-	-
Module – 2: GPU programming, Programming hybrid systems, MIMD systems, GPUs, Performance				
2.1	Speedup and efficiency in MIMD systems	2	-	-
2.2	Amdahl's law	2	-	-
2.3	Scalability in MIMD systems	2	-	-
2.4	Taking timings of MIMD programs	1	-	-
2.5	GPU performance	1	-	-
Module – 3: Distributed memory programming with MPI				
3.1	MPI functions	2	-	2
3.2	The trapezoidal rule in MPI	1	-	-
3.3	Dealing with I/O, Collective communication	2	-	1
3.4	MPI-derived datatypes, Performance evaluation of MPI programs	2	-	1
3.5	A parallel sorting algorithm	1	-	1
Module – 4: Shared-memory programming with OpenMP				
4.1	OpenMP pragmas and directives	2	-	2
4.2	The trapezoidal rule	1	-	-
4.3	Scope of variables, The reduction clause	1	-	-
4.4	Loop carried dependency, scheduling, producers and consumers	2	-	2
4.5	Caches, cache coherence and false sharing in OpenMP, tasking, tasking, thread safety.	2	-	1
Module – 5: GPU programming with CUDA				
5.1	GPUs and GPGPU, GPU architectures	1	-	-
5.2	Heterogeneous computing, Threads, blocks, and grids	2	-	-
5.3	Nvidia compute capabilities and device architectures, Vector addition	2	-	-
5.4	Returning results from CUDA kernels, CUDA trapezoidal rule I,	2	-	-
5.5	CUDA trapezoidal rule II: improving performance, CUDA trapezoidal rule III: blocks with more than one warp	1	-	-
Total No. of Lecture Hours		40	-	-
Total No. of Tutorial Hours			00	-
Total No. of Practical Hours				10

Practical Components	
Sl. No.	Experiments
1	Write an OpenMP program to print - Hello World - Thread (Process) Number and Total Number of Threads (Processes) - Parallel Addition on array of N Numbers - Find Minimum and Maximum Element in an Array
2	Write an OpenMP program to - Parallel Array Summation Using OpenMP with Partial Sums and Atomic Update - demonstrate Section Directives
3	Write an OpenMP program that divides the Iterations into chunks using various scheduling techniques. For example , if there are two threads and four iterations, the output might be the following: - Thread 0 : Iterations 0 – 1 - Thread 1 : Iterations 2 – 3
4	Write a OpenMP program to calculate n Fibonacci numbers using tasks.
5	Write a OpenMP program to find the prime numbers from 1 to n employing parallel for directive. Record both serial and parallel execution times
6	Write a MPI Program to demonstration of - MPI_Send and MPI_Recv - MPI_Send() and MPI_Recv() for Addition of Numbers
7	Write a MPI program to demonstration of deadlock using point to point communication and avoidance of deadlock by altering the call sequence
8	Write a MPI Program to demonstration of Broadcast operation.
9	Write a MPI Program demonstration of MPI_Scatter and MPI_Gather
10	Write a MPI Program to demonstration of MPI_Reduce and MPI_Allreduce (MPI_MAX, MPI_MIN, MPI_SUM, MPI_PROD)

Textbook:

1. Textbook 1

Peter S Pacheco, Matthew Malensek

An Introduction to Parallel Programming, Second edition, Elsevier Morgan Kauffman publications

Reference Book:

1. Michael J Quinn – Parallel Programming in C with MPI and OpenMp, McGrawHill